Solving Equations with Variables on Both Sides
Answer Key

Grade 8 / Pre-Algebra

Quick Answers

1. 4
2. 3
3. 5
4. 6
5. 7
6. (a) the simplified sides $= 2x + 6$, —
7. -7
8. 4 hours
9. (a) the solution set $= \emptyset$, —
10. (a) the correct solution $= 3$, —

What is verified

7 of 10 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 6, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra
8.EE.C.7 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 13 tagged checks.

1. Balance scale: 3 bags and 4 blocks on the left, one bag and 12 blocks on the right. Write and solve the equation.
Solution: Let x be the number of blocks in one bag. The pans balance, so $3x + 4 = x + 12$. Take one bag off *both* pans (the balance is kept), leaving $2x + 4 = 12$; take four blocks off both pans to get $2x = 8$; then halve both pans. $\boxed{x = 4}$

2. Solve $5x = 2x + 9$.
Solution: Subtract $2x$ from both sides to leave $3x = 9$, then divide both sides by three. $\boxed{x = 3}$

3. Solve $7x - 5 = 4x + 10$.
Solution: Subtract $4x$ from both sides: $3x - 5 = 10$. Add five to both sides (the inverse of subtracting): $3x = 15$. Divide by three. $\boxed{x = 5}$

4. Gym A: 20 dollars plus 3 dollars a class. Gym B: 8 dollars plus 5 dollars a class. When is the cost equal?
Solution: Let n be the number of classes. The two totals are equal when $20 + 3n = 8 + 5n$. Subtract $3n$ from both sides: $20 = 8 + 2n$. Subtract eight: $12 = 2n$. Halve both sides. Six classes is a sensible answer in the story — a whole number of classes. $\boxed{n = 6}$

5. Solve $4(x-2) = 2x+6$.

Solution: Clear the bracket first: $4x - 8 = 2x + 6$. Subtract $2x$: $2x - 8 = 6$. Add eight: $2x = 14$. Halve. $\boxed{x = 7}$

6. $2(x+3) = 2x+6$: (a) clear the bracket, (b) how many solutions?

Solution (a): Distributing on the left multiplies both terms inside the bracket, and the two sides turn out identical. $\boxed{2x + 6 = 2x + 6}$

Solution (b) — instructor-judged. Because the two sides are the same expression, the equation is true whatever number is substituted, so every real number is a solution — infinitely many. Full credit requires that identity observation, not just the words "infinitely many" on their own.

7. Solve $3(2x+1) = 5x-4$.

Solution: Clear the bracket: $6x+3 = 5x-4$. Subtract $5x$ from both sides: $x+3 = -4$. Subtract three from both sides. A negative solution is perfectly legitimate here; substituting it back gives -39 on both sides. $\boxed{x = -7}$

8. Candles: 24 cm burning 3 cm an hour, and 16 cm burning one cm an hour. When are they equal?

Solution: Let t be the number of hours. Each height is its start minus its burn rate times the time, so $24 - 3t = 16 - t$. Add $3t$ to both sides: $24 = 16 + 2t$. Subtract sixteen: $8 = 2t$. Halve. $\boxed{t = 4 \text{ hours}}$

9. $5x+3 = 5x-2$: (a) collect the variable terms, (b) how many solutions?

Solution (a): Subtracting $5x$ from both sides removes the variable altogether and leaves the numerical claim $3 = -2$. $\boxed{3 = -2}$

Solution (b) — instructor-judged. What is left is false: three is not negative two. No number was involved in that statement, so no value of the variable can rescue it, and the equation has $\boxed{\text{no solution}}$

. Full credit requires the leftover false statement to be named, not just the words "no solution".

10. Challenge. Elena's work on $7x - 6 = 2x + 9$. (a) Solve correctly. (b) What went wrong?

Solution (a): Subtract $2x$ from both sides: $5x - 6 = 9$. Now *add* six to both sides — adding is the inverse of subtracting — to get $5x = 15$, then divide by five. $\boxed{x = 3}$

Solution (b) — instructor-judged. Elena's first line is right. Her second line subtracts six from both sides when the six had to be added: she repeated the operation she saw instead of applying its inverse. The substitution check makes it plain — her value does not make the two sides of the original equation equal, while the correct one gives fifteen on both sides. Full credit requires naming the inverse-operation rule.